

EDUCATION

School of Informatics, The University of Edinburgh

Edinburgh, UK

PhD in NLP; 2026 Apple PhD Scholar in AI/ML

June 2027 expected

Beihang University

Beijing, China

Master of Electronic Information; GPA: 3.9/4.0; Ranking: 2/86

June 2023

Beihang University

Beijing, China

Bachelor of Information and Computing Science; GPA: 3.7/4.0; Ranking: 6/87

June 2020

EXPERIENCE

AliStar Research Intern, Qwen Pretraining Team

Jul. 2026 - Present

Advisors: Dr. Bo Zheng & Zihan Qiu

Beijing

TopSeed Research Intern, ByteDance Seed Speech Team

Mar. 2026 - Jul. 2026

Advisors: Dr. Wenhao Zhu & Shanbo Cheng

Shanghai

Student Researcher, Google DeepMind Continual Learning Team

Aug. 2025 - Feb. 2026

Advisors: Dr. Marc'Aurelio Ranzato & Dr. Adhiguna Kuncoro

London

Research Intern at INF Technology

Mar. 2024 - Aug. 2024

Advisor: Zili Wang

Remote

Research Intern at BAAI

Sep. 2022 - Jun. 2023

Advisor: Dr. Jie Fu

Beijing, China

Research Intern at WeChat AI

Nov. 2021 - May. 2022

Advisor: Dr. Yikang Shen

Beijing, China

RESEARCH INTERESTS

I am broadly interested in how large language models learn, including (1) the learning paradigms we use: how models learn from corrections, demonstrations, rewards, and context; (2) the objectives we optimize: how training objectives shape their behavior; and (3) the model architectures we build: how modularity, routing, and attention affect learning.

PUBLICATIONS

* denotes equal contribution.

Zihan Qiu*, **Zeyu Huang***, Kaiyue Wen*, Peng Jin*, Bo Zheng*,..., Ivan Titov, Dayiheng Liu, Jingren Zhou, Junyang Lin. A Unified View of Attention and Residual Sinks: Outlier-Driven Rescaling is Essential for Transformer Training - **COLM 2026** **Oral**.

Zeyu Huang, Tianhao Cheng, Zihan Qiu, Zili Wang, Yinghui Xu, Edoardo M Ponti, Ivan Titov. Blending Supervised and Reinforcement Fine-Tuning with Prefix Sampling - **ICML 2026**

Yijun Yang*, **Zeyu Huang***, Wenhao Zhu, Zihan Qiu, Fei Yuan, Jeff Z Pan, Ivan Titov. A Controllable Examination for Long-Context Language Models - **NeurIPS 2025** **Spotlight**

Zihan Qiu*, Zekun Wang*, Bo Zheng*, **Zeyu Huang***, Kaiyue Wen, Songlin Yang, ... & Junyang Lin. Gated Attention for Large Language Models: Non-linearity, Sparsity, and Attention-Sink-Free. **NeurIPS 2025** **Oral & Best Paper Award**

Zihan Qiu*, **Zeyu Huang***, Bo Zheng, Kaiyue Wen, Zekun Wang, ... & Junyang Lin. Demons in the Detail: On Implementing Load Balancing Loss for Training Specialized Mixture-of-Expert Models - **ACL 2025**

Zeyu Huang, Zihan Qiu, Zili Wang, Edoardo M Ponti, Ivan Titov. Post-hoc Reward Calibration: A Case Study on Length Bias - **ICLR 2025**

Zihan Qiu*, **Zeyu Huang***, Shuang Cheng, Yizhi Zhou, Zili Wang, Ivan Titov, Jie Fu. Layerwise Recurrent Router for Mixture-of-Experts - **ICLR 2025**

Ka Man Lo*, **Zeyu Huang***, Zihan Qiu*, Zili Wang, Jie Fu. A Closer Look into Mixture-of-Experts in Large Language Models. **NAACL 2025 Findings**

Wenyu Du, Tongxu Luo, Zihan Qiu, **Zeyu Huang**, Yikang Shen, Reynold Cheng, Yike Guo, Jie Fu. Stacking Your Transformers: A Closer Look at Model Growth for Efficient LLM Pre-Training - **NeurIPS 2024** **Spotlight**.

Wenyu Du, Shuang Cheng, Tongxu Luo, Zihan Qiu, **Zeyu Huang**, Ka Chun Cheung, Reynold Cheng, Jie Fu. Unlocking Continual Learning Abilities in Language Models - **EMNLP 2024 Findings**

Zihan Qiu*, **Zeyu Huang***, Jie Fu. Unlocking Emergent Modularity in Large Language Models. **NAACL 2024**. **Outstanding Paper Award**

Zihan Qiu*, **Zeyu Huang***, Youcheng Huang*, Jie Fu. Empirical Study on Updating Key-Value Memories in Transformer Feed-forward Layers. Tiny Papers @ **ICLR 2024**

Zeyu Huang, Yikang Shen, Xiaofeng Zhang, Jie Zhou, Wenge Rong, Zhang Xiong. Transformer-Patcher: One Mistake Worth One Neuron. **ICLR 2023**.

Zeyu Huang, Wenge Rong, Xiaofeng Zhang, Yuanxin Ouyang, Chenghua Lin, Zhang Xiong. Token Relation Aware Chinese Named Entity Recognition. **ACM TALLIP**

Zeyu Huang, Xiaofeng Zhang, Jun Bai, Wenge Rong, Yuanxin Ouyang, Zhang Xiong. Solving Math Word Problems Following Logically Consistent Template. **IJCNN 2023**

Xiaofeng Zhang*, Yikang Shen*, **Zeyu Huang**, Jie Zhou, Wenge Rong, Zhang Xiong. Mixture of Attention Heads: Selecting Attention Heads Per Token. **EMNLP 2022**

PRE-PRINTS

Zeyu Huang, Adhiguna Kuncoro, Qixuan Feng, Jiajun Shen, Lucio Dery, Arthur Szlam, Marc'Aurelio Ranzato. Context Training with Active Information Seeking

Tianhao Cheng*, **Zeyu Huang***, Zihan Qiu, Yu Cheng, Edoardo Ponti, Yinghui Xu, Ivan Titov, Zenglin Xu. The Cancellation Hypothesis in Critic-Free RL: From Outcome Rewards to Token Credits

HONORS AND AWARDS

- 2026 Apple PhD Scholar in AI/ML (20 worldwide)
- NeurIPS 2025 Best Paper Award (4/21575 submissions)
- NAACL 2024 Outstanding Paper Award (6/2604 submissions)
- EPSRC DTA Scholarship (international flexibility) - 2024-2027
- National Scholarship for Graduate Students Award - 2021-2022 - (3/291)
- Outstanding Graduate Student of Beihang University - 2020-2021

EXTRA-CURRICULAR ACTIVITIES

International Exchange	Casablanca, Morocco
Ecole Centrale Casablanca	Nov 2019, 15 days

International Exchange	Paris, France
CentraleSupélec, Saclay	Feb 2018, 8 days

Student works	Beijing, China
President of the Graduate Student Union of ECPK	Dec. 2020 - Dec. 2021

Orange Lab & Ecole Centrale de Pekin	Beijing, China
Oct 2020 - May 2021	Group Leader of the Project

- **Federated Recommendation** - A recommendation system trained in Federated Learning manner.
- This project is co-supervised by Orange Lab of Beijing and Ecole Centrale de Pekin.
- The product of this project includes a survey report and deployment of a Federated news recommender

SKILLS SUMMARY

- **Programming:** Python, C, Shell, JAVA, Latex, MATLAB
- **Languages:** Chinese(Native), English (IELTS 7.5), French (B2)
- **ML/AI Frameworks:** PyTorch, JAX, TensorFlow, PyTorch Lightning, Fairseq, Hugging Face, Flax